AMATH 482/582: HOME WORK 1

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Abstract. This report aims to solve the problem of reconstructing a clear path of a submarine. A dataset containing containing 49 measurements of its acoustic pressure over the span of 24 hours is provided. This report is split into 5 sections. The introduction, which outlines the problem statement. The theoretical background which showcases the necessary formulas and concepts necessary to solve the problem. The algorithm implementation, which shows which libraries and algorithms were used, the summarization of my computational results, and a conclusion of my findings. Ultimately I denoised the provided measurements in the data by finding the dominant frequency using the average of the Fourier transformations and creating a Gaussian filter using the dominant frequency. I then used the Fast Fourier Transformation method to convert each measurement to its frequencies, applied the Gaussian filter to remove the noise, then used the Inverse Fast Fourier Transformation method to convert it back to its measurement showing a clear location.

1. Introduction and Overview

The goal of this homework is to locate a submarine that is moving in the Puget Sound. We do not know much about this submarine as it is a new technology that emits an unknown acoustic frequency that we must detect. Only broad spectrum recording of acoustics pressure data obtained over 24 hours in half-hour increments is available.

2. Theoretical Background

The goal of this assignment can be split into 3 different parts: 1. Obtain the frequencies of each signal. 2. Filter out the noise in the frequencies of the signals. 3. Convert the frequencies back into denoised signals.

We know that adding mean zero white noise to a signal (Gaussian Noise) is equivalent to adding mean zero white noise (Gaussian Noise) to its coefficients. Thus, we can reduce the white noise by averaging the fourier coefficients and finding the dominant frequency. The dominant frequency can be obtained through the averaging of the Fourier transform.

This will give us a formula like $\frac{\sum_{i=1}^{n} |F(x_i, y_i, z_i)|}{n}$ where $|F(x_i, y_i, z_i)|$ is the Fourier transform of signal i and $n = 49$.

For the second part, we want to use our dominant frequency to create the 3-dimensional Gaussian filter. We can use the formula: $G(k_x, k_y, k_z) = e^{-T((k_x-k_x^*)^2+(k_y-k_y^*)^2+(k_z-k_z^*)^2}$ where $T = \frac{1}{2\sigma^2}$ (k_x, k_y, k_z) is every point on the grid and (k_x^*, k_y^*, k_z^*) is the point of the dominant frequency. We are essentially calculating the distance between these points and the center/dominant point. The magnitude of σ determines how much noise in a frequency of a signal gets suppressed.

Now we finally go through each signal and suppress the noise. We do this by applying the Fourier transform to turn the signal into frequencies, applying the filter to remove the noise from

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the frequencies, then applying the inverse Fourier transform to turn the frequencies back into a signal.

In mathematical form this looks like: $F^{-1}(G(k_x, k_y, k_z) * F(x_i, y_i, z_i)$ for i in 1,...,n where $n = 49$.

3. Algorithm Implementation and Development

I used numpy for several numerical operations [1]. For example I used the np.fft.fftn and np.fft.fftshift to compute the Fourier shift on the data. I used unravel_index to convert flat arrays into 3D coordinates (for example I converted the index of the max Fourier transform into 3D coordinates). I used argmax to find the maximum value in an array (I used it to find the dominant frequency and track the submarine's location as it finds the max intensity at any time). I used Plotly (go.Isosurface and go.Scatter3d) to generate 3D plots [3]. Finally I used Matplotlib to generate a 2D visualization of the submarine's path [2].

4. Computational Results

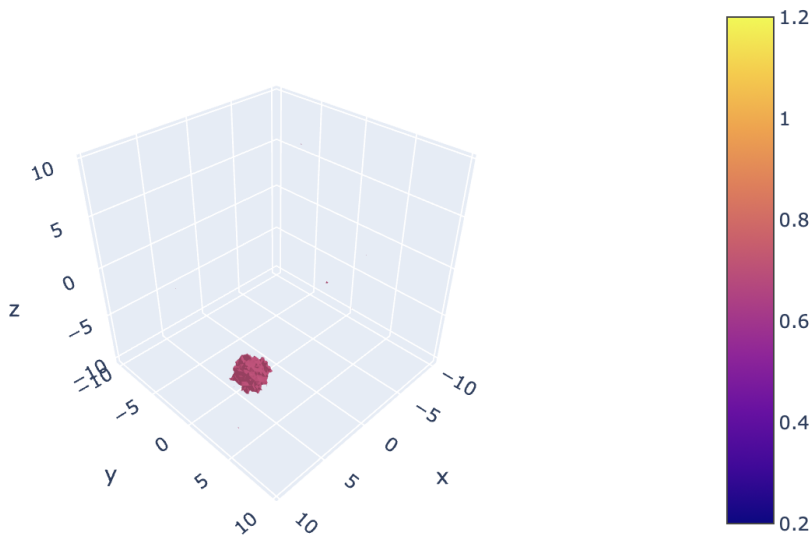

Figure 1. [3] This figure showcases the dominant frequency created by the average of the Fourier transformations of each original signal. You can observe that the extra noise seen in the original path of the submarine is missing and the points are clumped around each other. The location of the dominant frequency is at (1.89, 5.34, -6.91).

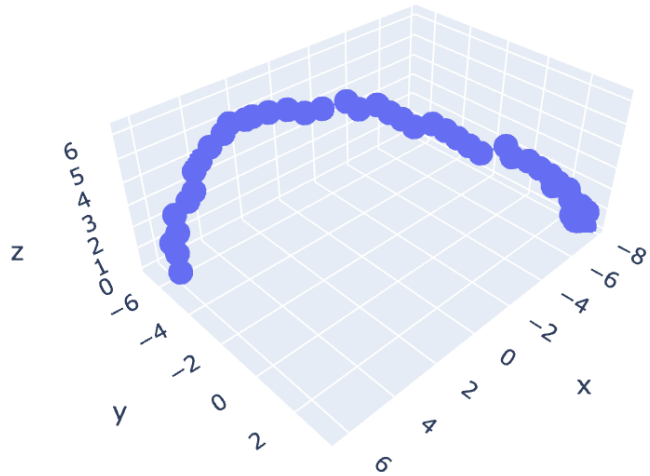

FIGURE 2. [3] This is the visualization of the submarine path after each signal was converted to its frequencies, the filter created using the dominant frequency was applied onto these frequencies, and the denoised frequencies were then converted back to signals. You can see that the location of the submarine is much clearer in this 3D path than the original 3D path.

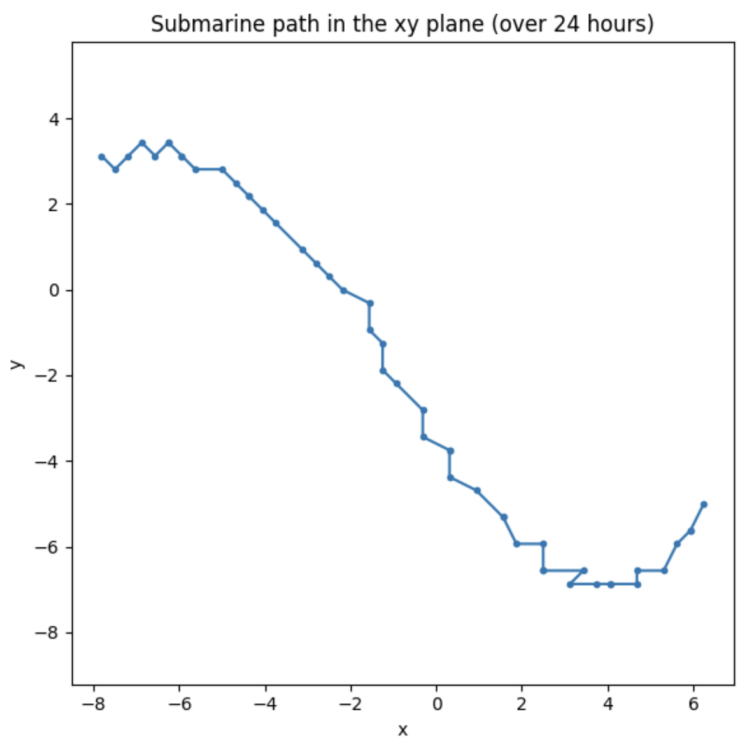

FIGURE 3. [2] In this figure we can see the reconstructed path of the submarine in the XY plane. This path looks smooth and continuous, allowing us to observe the exact (x,y) location of the submarine. This figure allows us to pinpoint the exact (x,y) location of the submarine at any hour over the 24 hour period.

5. Summary and Conclusions

In this report I reconstructed a clear path of a submarine using a dataset containing 49 measurements of its acoustic pressure over the span of 24 hours. I did this by finding the dominant acoustic frequency by taking the average of the Fourier transformations, and then using this dominant frequency to create a Gaussian filter. I then converted each measurement to its frequencies using the Fast Fourier Transformation method, applied the Gaussian filter to remove the noise, then used the Inverse Fast Fourier Transformation method to convert it back to its measurement showing a clear location. I then presented results showing both the dominant frequency and the reconstructed path in 2D and 3D. Some future directions could be building an algorithm that can take in any set of noisy signals and output the denoised signals.

Acknowledgements

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References

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